MILITARY SPECIFICATION SHEET

ATTENUATORS, VARIABLE (COAXIAL),
CLASS 1 (N)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the attenuators described herein shall consist of this document and the latest issue of Specification MIL-A-24215.

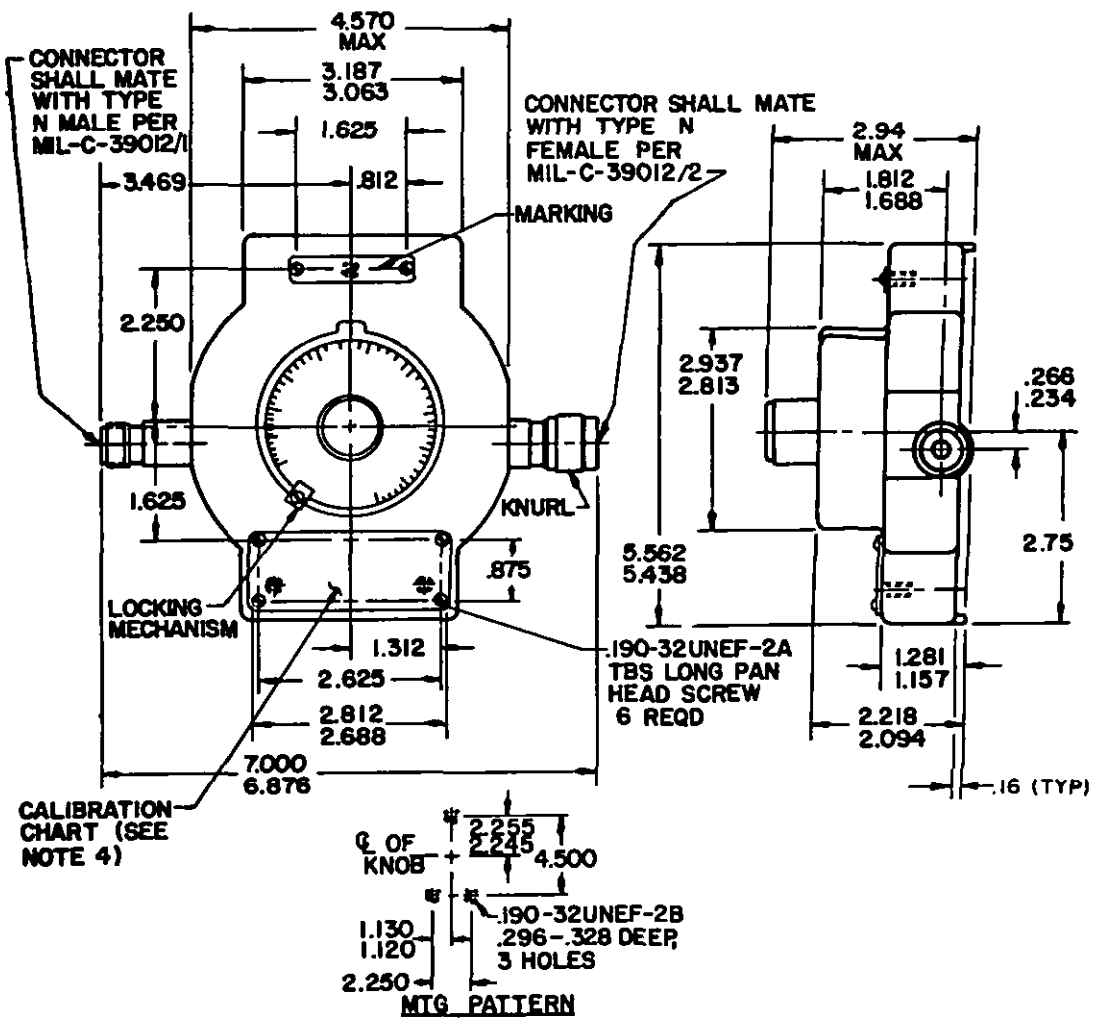


FIGURE 1. Dimensions and configuration for part number M24215/12-001.

FSC 5985

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.16	4.07	1.120	28.45	2.094	53.19	2.812	71.42	4.500	114.30
.190	4.83	1.130	28.70	2.218	56.34	2.813	71.45	4.570	116.08
.234	5.94	1.157	29.39	2.245	57.02	2.937	74.60	5.438	138.13
.266	6.76	1.281	32.54	2.250	57.15	2.94	74.68	5.562	141.27
.296	7.52	1.312	33.32	2.255	57.28	3.063	77.80	6.876	174.65
.328	8.33	1.625	41.28	2.625	66.68	3.187	80.95	7.000	177.80
.812	20.62	1.688	42.88	2.688	68.28	3.469	88.11		
.875	22.23	1.812	46.02	2.75	69.85				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.02 (.51 mm) for two place decimals.
4. Calibration for insertion loss shall be provided at DC, 0.3, 0.5, 0.75, and 1.0 GHz for the 0, 3, 6, and 10 dB dial settings.
5. The clockwise rotation of the calibrated dial continuously increases attenuation from 0 to 10 dB. Dial marking at 1 dB and 0.2 dB increments. Resetability shall be within 0.2 dB at 0.5 GHz.

FIGURE 1. Dimensions and configuration for part number M24215/12-001 - Continued.

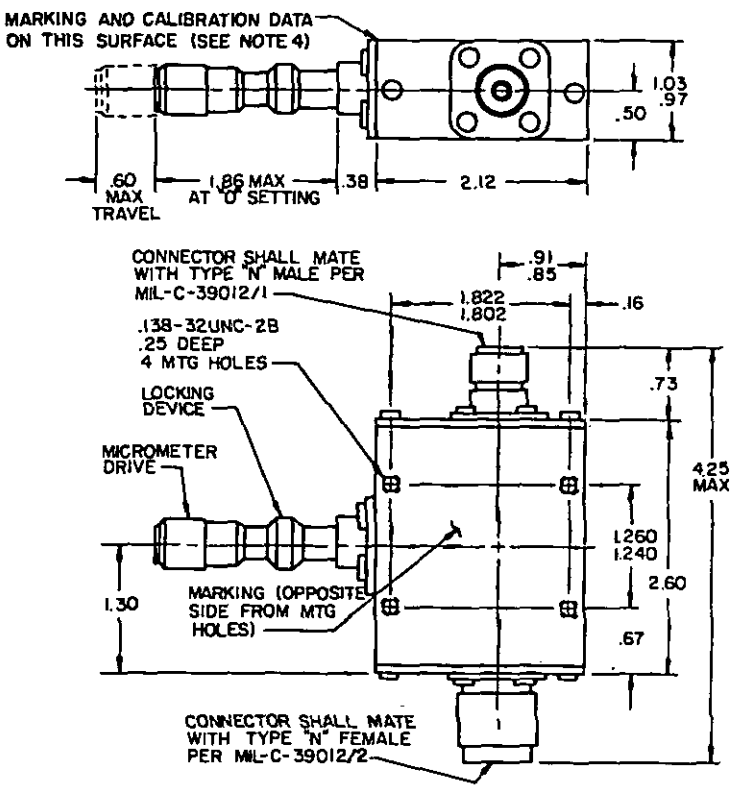


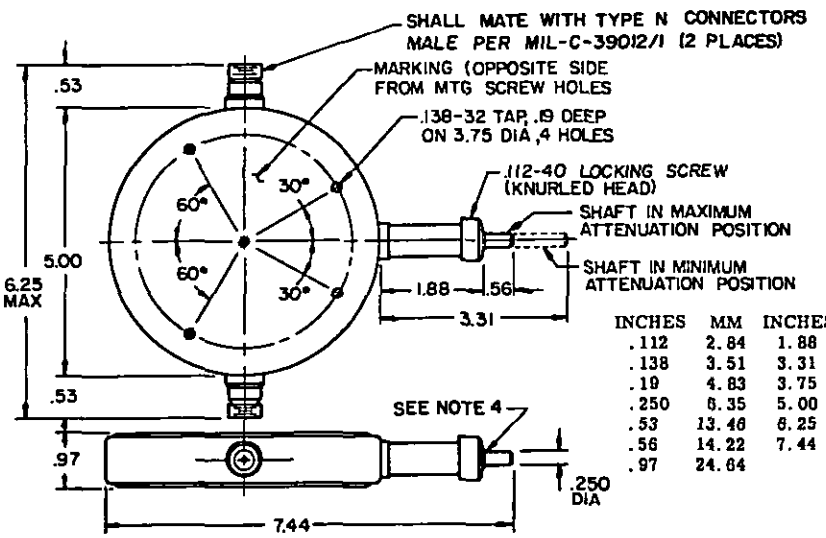
FIGURE 2. Dimensions and configuration for part number M24215/12-002.

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INCHES	MM	INCHES	MM	INCHES	MM
.138	3.51	.73	18.54	1.30	33.02
.16	4.06	.85	21.59	1.802	45.77
.25	6.35	.91	23.11	1.822	46.28
.38	9.65	.97	24.64	1.86	47.24
.50	12.70	1.03	26.16	2.12	53.85
.60	15.24	1.240	31.50	2.60	66.04
.67	17.02	1.260	32.00	4.25	107.95

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .02 (.51 mm).
4. Calibrate at 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0. and 10 dB settings in 100 MHz steps from 0.5 to 1.0 GHz.
5. A calibration chart, showing the relationship between the micrometer setting and the actual attenuation in dB, shall be furnished as permanent marking in the form of a permanently attached data plate.

FIGURE 2. Dimensions and configuration for part number M24215/12-002 - Continued.

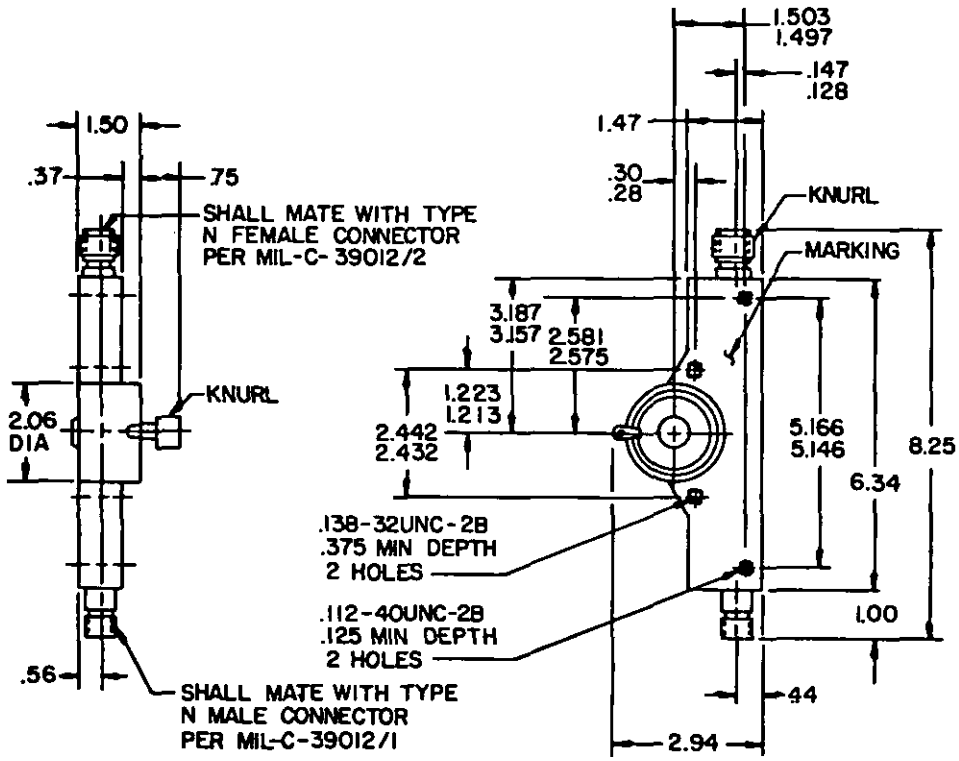


INCHES	MM	INCHES	MM
.112	2.84	1.88	47.75
.138	3.51	3.31	83.82
.19	4.83	3.75	95.25
.250	6.35	5.00	127.00
.53	13.46	6.25	158.75
.56	14.22	7.44	188.98
.97	24.64		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
4. Approximately 20 revolutions of shaft required to move attenuation through full range. Clockwise rotation of shaft increases loss. Shaft shall be provided with mechanical stops at limits.

FIGURE 3. Dimensions and configuration for part number M24215/12-003.

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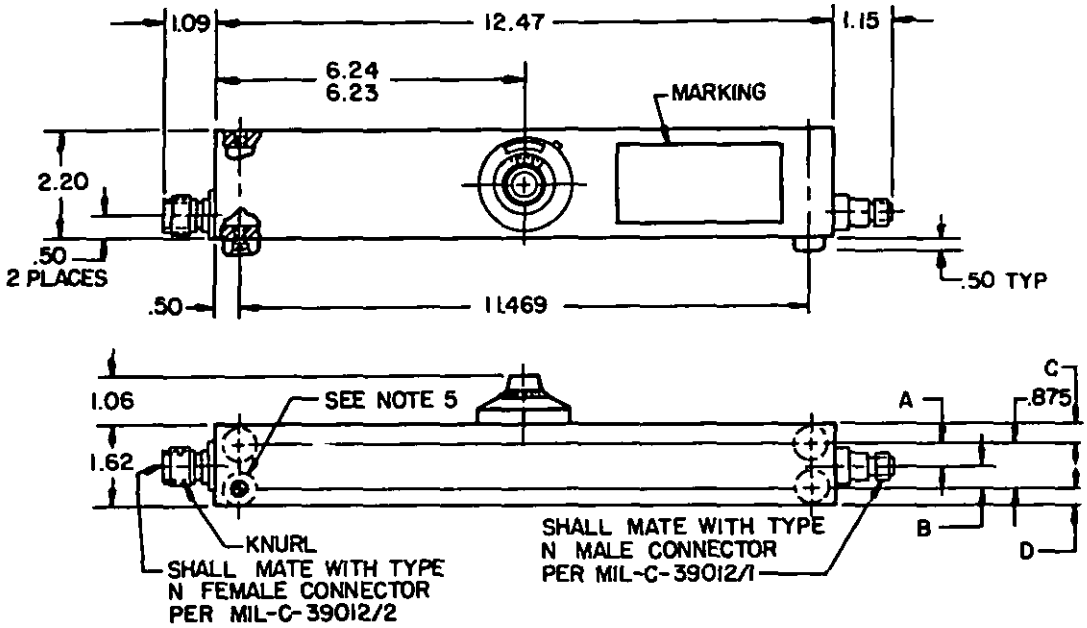


INCHES	MM	INCHES	MM	INCHES	MM
.112	2.84	.56	14.22	2.442	62.03
.125	3.18	.75	19.05	2.575	65.41
.128	3.25	1.00	25.40	2.581	65.56
.138	3.51	1.213	30.81	2.94	74.68
.147	3.73	1.223	31.06	3.157	80.19
.28	7.11	1.47	37.34	3.187	80.95
.30	7.62	1.497	38.02	5.146	130.71
.37	9.40	1.50	38.10	5.166	131.22
.375	9.53	1.503	38.11	6.34	161.04
.44	11.18	2.06	52.32	8.25	209.55
		2.432	61.77		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
4. A locking device shall be provided (as a part of the unit) which is capable of locking at any attenuation setting.

FIGURE 4. Dimensions and configuration for part number M24215/12-004.

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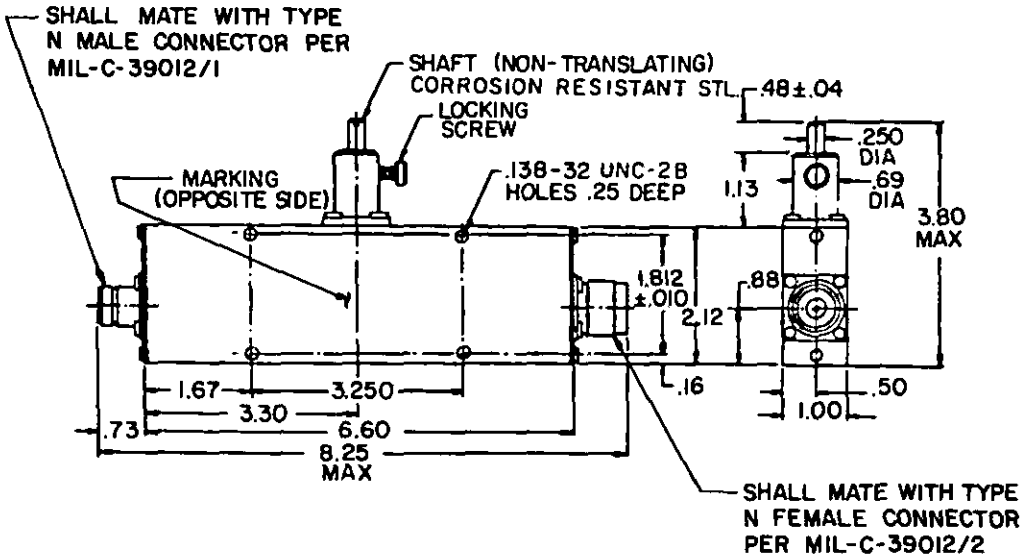


INCHES	MM	INCHES	MM
.50	12.70	1.62	41.15
.875	22.23	2.20	55.88
1.06	26.92	6.23	158.24
1.09	27.69	6.24	158.50
1.15	29.21	11.469	291.31
		12.47	316.74

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. A and B to be equal within .030 (.76 mm).
4. C and D to be equal within .030 (.76 mm).
5. Feet may be removed to use .138 (3.51 mm) -32 screws for mounting, .250 (6.35 mm) max thread insertion in housing, 4 places on both sides.
6. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
7. A locking device shall be provided (as a part of the unit) which is capable of locking at any attenuation setting.

FIGURE 5. Dimensions and configuration for part number M24215/12-005.

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Check the source to verify that this is the current version before use.

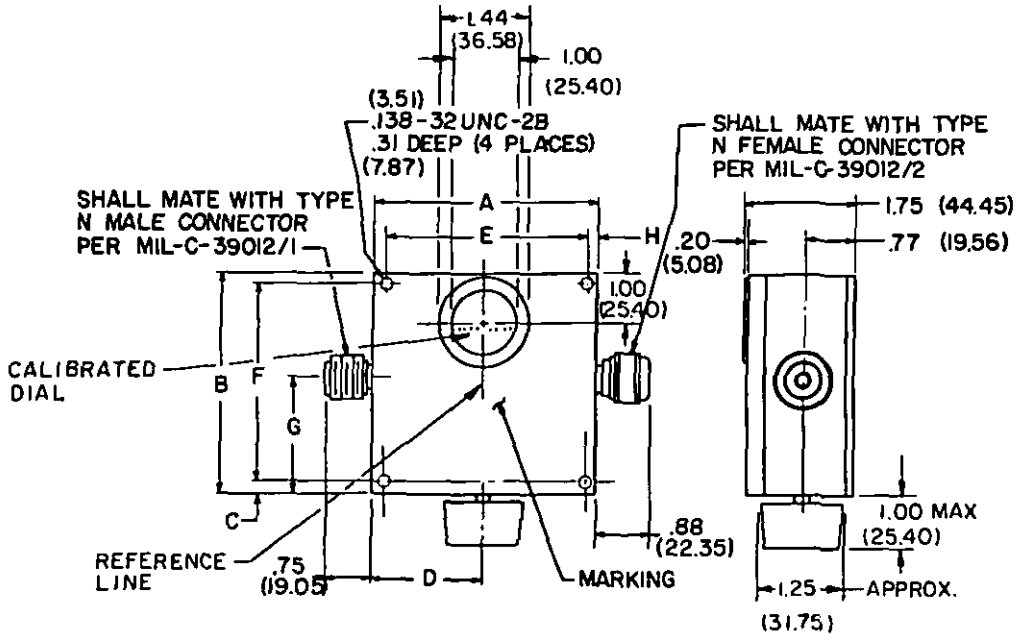


INCHES	MM	INCHES	MM	INCHES	MM
.010	.25	.50	12.70	1.812	46.02
.03	.76	.69	17.53	2.12	53.85
.04	1.02	.73	18.54	3.250	82.55
.138	3.51	.88	22.35	3.30	83.82
.16	4.06	1.00	25.40	3.80	96.52
.25	6.35	1.13	28.70	6.60	167.64
.48	12.19	1.67	42.42	8.25	209.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.

FIGURE 6. Dimensions and configuration for part number M24215/12-006.

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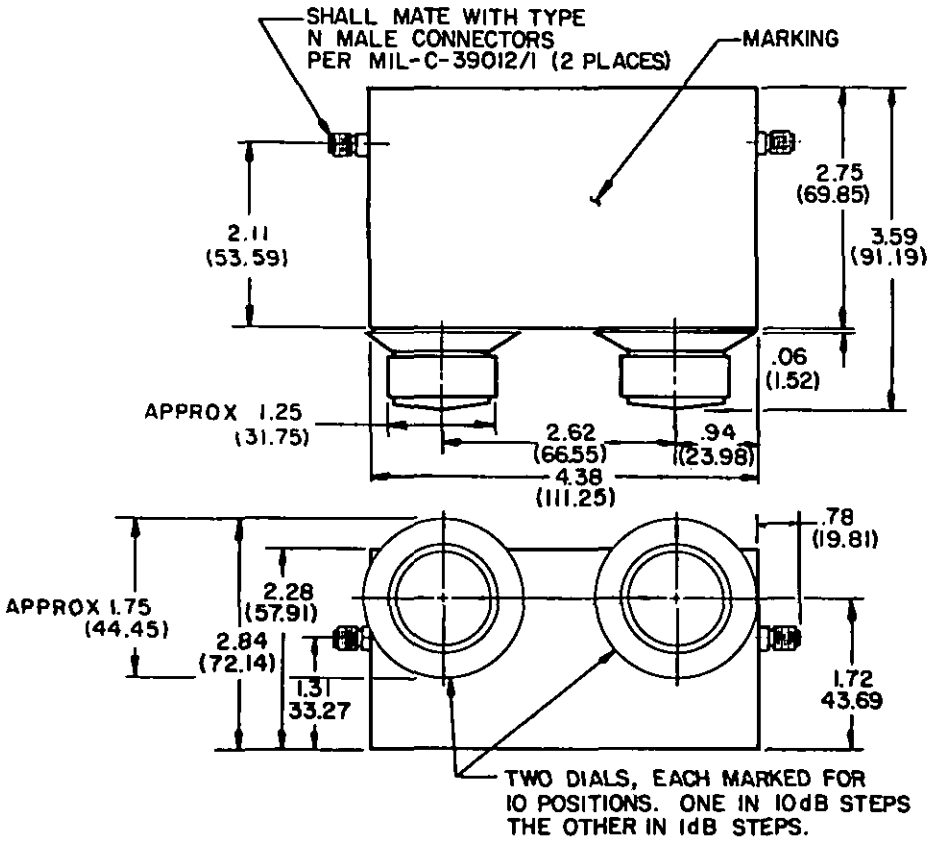


Part No. M24215 /12	A	B	C	D	E	F	G	H
007	4.83 (122.68)	4.28 (108.71)	.156 (3.96)	2.42 (61.47)	4.500 (114.30)	3.969 (100.81)	2.23 (56.64)	.156 (3.96)
008	8.50 (215.90)	5.23 (132.84)	.156 (3.96)	4.25 (107.95)	6.938 (176.23)	4.938 (125.43)	2.70 (68.58)	.781 (19.84)
009	6.75 (171.45)	4.61 (117.09)	.469 (11.91)	3.38 (85.85)	6.156 (156.36)	3.672 (93.27)	2.41 (61.21)	.297 (7.54)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.

FIGURE 7. Dimensions and configuration for part number M24215/12-007, 008 and 009.

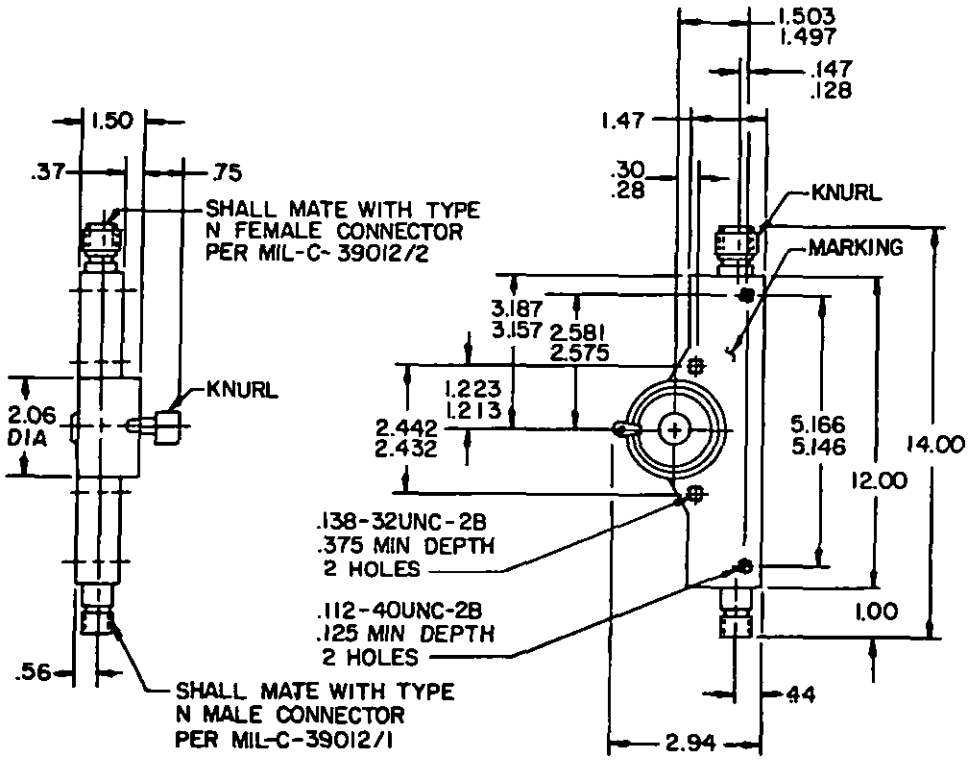
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerance is ±.03 (.76 mm).
5. A calibration chart, showing the actual attenuation for eacn step shall be furnished as permanent marking in the form of a permanently attached data plate.

FIGURE 8. Dimensions and configuration for part number M24215/12-010.

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INCHES	MM	INCHES	MM	INCHES	MM
.112	2.84	.56	14.22	2.442	62.03
.125	3.18	.75	19.05	2.575	65.41
.128	3.25	1.00	25.40	2.581	65.56
.138	3.51	1.213	30.81	2.94	74.68
.147	3.73	1.223	31.06	3.157	80.19
.28	7.11	1.47	37.34	3.187	80.95
.30	7.62	1.497	38.02	5.146	130.71
.37	9.40	1.50	38.10	5.166	131.22
.375	9.53	1.503	38.11	12.00	304.80
.44	11.18	2.06	52.32	14.00	355.60
		2.432	61.77		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
4. A locking device shall be provided (as a part of the unit) which is capable of locking at any attenuation setting.

FIGURE 8. Dimensions and configuration for part number M24215/12-011.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 through 8.

Nominal impedance: 50 ohms.

Electrical requirements: See table I.

Part number: See figures 1 through 8 and table I.

TABLE I. Electrical requirements. 1/ 2/

Part number M24215/12-	Figure	Freq. range (GHz)	Atten. range (dB)	VSWR (max)	Insertion loss (max at min. atten. setting) (dB)	Calibration accuracy (dB)	Frequency sensitivity (dB)	Max input (watts)		Nominal operating temp. range (°C)	Weight (max)		
								av.	pk.		lbs	g	oz
001	1	DC-1.0	0-10	1.5:1	0.8	0.2	1.0	2	60	0.0 - +65	2.5	(1134)	40
002	2	0.5-1.0	0-10	1.5:1	0.5	±0.2	-	2	60	0.0 - +65	1	(454)	16
003	3	.8-2.5	0-30	1.5:1	0.2	-	-	10	5 kW	0.0 - +65	1.75	(794)	28
004	4	3.0-3.7	0-10	1.5:1	0.5	-	-0.4	1	1 kW	0.0 - +65	2	(907)	32
005	5	2.0-5.0	0-20	1.5:1	0.5	-	0.2 or 5%	2	2 kW	0.0 - +65	2.5	(1134)	40
006	6	0.3-8.0	0-25 @ 0.3 GHz to 8.0 GHz	1.5:1	0.7	-	-	10	-	0.0 - +65	1.6	(726)	25
007	7	4.0-8.0	0-20	1.25:1	1.0	±1.0	-	10	5 kW	0.0 - +65	2.7	(1225)	44
008	7	4.0-8.0	0-40	1.25:1	1.5	±1.5	-	10	5 kW	0.0 - +65	5.2	(2359)	84
009	7	7.0-11.0	0-40	1.25:1	1.5	±1.5	-	10	5 kW	0.0 - +65	4.0	(1814)	64
010	8	DC-12.4	0-99	1.50:1	0.8	±2.0	-	2	200	0.0 - +65	2.7	(1225)	43
011	9	2.5-11.0	0-20	1.60:1 @ 2.5-4.0 GHz 1.50:1 @ 4-11 GHz	1.0	±0.2	0.5	1	1 kW	0.0 - +65	1.25	(568)	20

NOTES:
1/ Sweep frequency techniques may be used.
2/ Continuously variable except for M24215/12- 010 which is variable in 1 dB steps.

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NOTE: Attenuators covered by this specification sheet are substitutable for the following commercial types:

Part number M24215/12-	FSN 5985	Manufacturer's code	Commercial type
001	840-2394	93459	905
001		49956	713243-1
001	946-3183	93459	905S
001	107-2724	93459	905LS
001	134-6859	93459	2971
001	134-6859	18876	7913364
002		49956	580734-1
002		12457	AU-25A/VTS
003		09992	2-3416-30
004		10001	2902864-2
004		93459	953PLSL-10
004		93459	953P10
004		93459	953PL10
004		12598	A2091
005		10001	2902864-1
005		12598	A2078
005		12598	AV-35-N
006		12457	AU-65-A
007	135-4058	99899	793FM
008	069-1648	99899	794FM
009		99899	796FM
010	099-4475	99899	704-99
011		93459	973P-20

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, SG, MU
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - SM
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Navy - EC

Agent:
SA - ES

(Project 5985-0826)

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